Practice paper 2 – Exercise 1

Integration and L^p spaces | 7 points | Suggested time: 20–25 minutes

Problem statement. Show your calculations and give brief reasons. For $n \geq 1$, let $f_n(x) = x^n$ on $[0, 1]$, equipped with Lebesgue measure.

(a) **[2 points]** Determine $\lim_{n\to\infty} f_n(x)$ separately for $0 \leq x < 1$ and for $x = 1$. Is the limit zero almost everywhere?

(b) **[3 points]** Compute $\int_0^1 f_n(x)\,dx$, $\|f_n\|_{L^1}$, and $\|f_n\|_{L^2}$. Does $f_n \to 0$ in L^1 and in L^2?
Hint: $\int_0^1 x^a\,dx = 1/(a+1)$ for $a > -1$.

(c) **[2 points]** Verify that the constant function $g(x) = 1$ is an integrable bound for $|f_n|$. Apply the dominated convergence theorem to find the limit of the integrals.

Practice paper 2 – Solution 1

Detailed worked solution

(a) Pointwise and almost-everywhere limits. If $0 \le x < 1$, the powers x^n tend to zero. At $x = 1$, however, $f_n(1) = 1$ for every n. The pointwise limit is

$$f(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

The singleton $\{1\}$ has Lebesgue measure zero. Therefore $f = 0$ almost everywhere, and $f_n \to 0$ almost everywhere.

(b) Integrals and norms. The power integration formula gives

$$\int_0^1 f_n(x)\,dx = \int_0^1 x^n\,dx = \left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}.$$

As $x^n \ge 0$, $\|f_n\|_{L^1} = 1/(n+1) \to 0$. For L^2,

$$\|f_n\|_{L^2}^2 = \int_0^1 x^{2n}\,dx = \frac{1}{2n+1}, \qquad \|f_n\|_{L^2} = \frac{1}{\sqrt{2n+1}} \longrightarrow 0.$$

Thus $\boxed{f_n \to 0 \text{ in both } L^1[0,1] \text{ and } L^2[0,1].}$ The value at $x = 1$ does not affect either integral or norm: these quantities ignore changes on sets of measure zero.

(c) Dominated convergence. On $[0,1]$, $0 \le x^n \le 1$. The constant function $g = 1$ is measurable and has integral one. Each f_n is measurable, and part (a) gives almost-everywhere convergence to zero. Hence all hypotheses of dominated convergence are satisfied, and

$$\boxed{\lim_{n\to\infty} \int_0^1 x^n\,dx = \int_0^1 0\,dx = 0.}$$

One may equivalently use the exact pointwise limit f: its integral is also zero because its only nonzero value occurs on a singleton.

Key point. Almost-everywhere convergence is enough for the dominated convergence theorem. Convergence at every point is not required.

Practice paper 2 – Exercise 2

Probability and conditioning | 7 points | Suggested time: 20–25 minutes

Problem statement. Show your calculations and give brief reasons. Suppose the joint density of X and Y is constant on a rectangle:

$$f_{X,Y}(x,y) = \begin{cases} 1/2, & 0 < x < 1,\ 0 < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) **[2 points]** Compute the marginal densities f_X and f_Y, including their supports. Are X and Y independent?

(b) **[3 points]** Find $f_{Y|X=x}(y)$ for $0 < x < 1$, then compute $\mathbb{E}[Y \mid X = x]$.
Recall: $f_{Y|X=x}(y) = f_{X,Y}(x,y)/f_X(x)$ when $f_X(x) > 0$.

(c) **[2 points]** Compute $\mathbb{P}(X < 1/2, Y < 1)$ and $\mathbb{E}[Y]$.

Practice paper 2 – Solution 2

Detailed worked solution

(a) Marginal densities. For a fixed $x \in (0,1)$, integrate over $0 < y < 2$:

$$f_X(x) = \int_0^2 \frac{1}{2}\, dy = 1.$$

For a fixed $y \in (0,2)$, integrate over $0 < x < 1$:

$$f_Y(y) = \int_0^1 \frac{1}{2}\, dx = \frac{1}{2}.$$

Thus X is uniform on $(0,1)$ and Y is uniform on $(0,2)$. Each density is zero outside its stated interval.

On the rectangle, $f_X(x) f_Y(y) = 1 \cdot (1/2) = f_{X,Y}(x,y)$; outside it, both the product and the joint density are zero. Hence $\boxed{X \text{ and } Y \text{ are independent.}}$

(b) Conditional density and mean. For $0 < x < 1$, divide the joint density by $f_X(x) = 1$:

$$f_{Y|X=x}(y) = \begin{cases} 1/2, & 0 < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

The conditional density is unchanged by knowing $X = x$. Integrating against it gives

$$\mathbb{E}[Y \mid X = x] = \int_0^2 y\, \frac{1}{2}\, dy = \frac{1}{2}\left[\frac{y^2}{2}\right]_0^2 = 1.$$

Therefore $\mathbb{E}[Y \mid X] = 1$ almost surely.

(c) Joint probability and expectation. The event is a rectangle of side lengths $1/2$ and 1. Its probability is its area times the constant density:

$$\mathbb{P}(X < 1/2, Y < 1) = \int_0^{1/2} \int_0^1 \frac{1}{2}\, dy\, dx = \frac{1}{2} \cdot 1 \cdot \frac{1}{2} = \boxed{\frac{1}{4}}.$$

The tower property yields $\mathbb{E}[Y] = \mathbb{E}[1] = \boxed{1}$. Directly,

$$\mathbb{E}[Y] = \int_0^2 y f_Y(y)\, dy = \int_0^2 \frac{y}{2}\, dy = 1.$$

Key point. Under independence, conditioning on X does not change the distribution of Y.